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### SEMICONDUCTOR FABRICATION SYSTEM AND METHOD

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#### Abstract

A system and computer-implemented method are provided for manufacturing a semiconductor electronic device. An assembler receives a jig and a boat supporting a die. The assembler includes a separator that separates the jig into a first jig portion and a second jig portion and a loader that positions the boat between the first jig portion and the second jig portion. A robot receives an assembly prepared by the assembler and manipulates a locking system that fixes an alignment of the boat relative to the first jig portion and the second jig portion to form a locked assembly. A process chamber receives the locked assembly and subjects the locked assembly to a fabrication operation.

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## **Background/Summary**

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of and claims priority to U.S. Non-Provisional application Ser. No. 18/601,423, filed Mar. 11, 2024, which is a divisional of and claims the benefit of U.S. Non-Provisional application Ser. No. 16/939,820, filed Jul. 27, 2020, which claims the benefit of U.S. Provisional Application No. 62/894,303, filed Aug. 30, 2019. U.S. Non-Provisional application Ser. No. 18/601,423, U.S. Non-Provisional application Ser. No. 16/939,820, and U.S. Provisional Application No. 62/894,303 are incorporated in their entireties herein by reference.

### **BACKGROUND**

[0002] Semiconductor fabrication processes, such as very-large-scale integration (VLSI), create integrated circuits by forming networks of transistors and other circuit components on a single chip, commonly referred to as a die. Several dies can be formed on a single wafer, or dies can be processed individually after being cut from a wafer. Due to the number of circuit components included in modern integrated circuits, the fabrication processes have become increasingly complex. Dies, whether formed as part of a wafer or having been cut from a wafer, are transported to various different treatment chambers, process tools, and fabrication bays to be subjected to a sequence of processing operations.

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## **Description**

### **BRIEF DESCRIPTION OF THE DRAWINGS**

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1 illustrates a system for manufacturing a semiconductor electronic device, in accordance with some embodiments.

[0005] FIG. 2 is a perspective view of a jig, in accordance with some embodiments.

[0006] FIG. 3 is a top view of a boat supporting a plurality of dies, in accordance with some embodiments.

[0007] FIG. 4 is a perspective view of a boat magazine, in accordance with some embodiments.

[0008] FIG. 5 is a perspective view of an assembly comprising a boat between jig portions, in accordance with some embodiments.

[0009] FIG. 6 is a perspective view of an assembler separating a second jig portion from a first jig portion to form an assembly of jig portions and a boat supporting a die, in accordance with some embodiments.

[0010] FIG. 7 is a perspective view of a transport system, in accordance with some embodiments.

[0011] FIG. 8 is a top view of a system for manufacturing a semiconductor electronic device, in accordance with some embodiments.

[0012] FIG. 9 is a flow chart illustrating a method of manufacturing a semiconductor electronic

device, in accordance with some embodiments.

[0013] FIG. **10** illustrates an example computer-readable medium wherein processor-executable instructions configured to embody one or more of the provisions set forth herein may be comprised, in accordance with some embodiments.

[0014] FIG. **11** illustrates an example computing environment wherein one or more of the provisions set forth herein may be implemented, in accordance with some embodiments.

#### DETAILED DESCRIPTION

[0015] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying Figs. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0016] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments or configurations discussed.

[0017] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the Figs. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the Figs. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0018] According to some embodiments, a system and process for manufacturing a semiconductor device are provided. The system comprises an assembler that receives a jig and a boat supporting at least one die and forms an assembly, which is secured by a first robot. According to some embodiments, the jig is retrieved by a second robot from a jig carrier storing an inventory of available jigs. For example, the jig carrier comprises an array of shelves arranged in rows and columns. Each storage cell, corresponding to a unique intersection of row and column values, can store a jig, and information about the jig stored at each intersection or location can be maintained in a computer-accessible database. In some embodiments, the second robot comprises a robotic arm having a plurality of articulations allowing multi-axis movement. An example of the second robot includes an arm that is adjustable relative to 6-axes, allowing movement along or relative to each of the 6 axes.

[0019] The boat supporting at least one semiconductor die is supplied to and received by the assembler. For example, the boat, supporting the die comprising electronic circuitry that is to be subjected to a fabrication step, can be introduced to the system through an entry load port. In some embodiments, a conveyor is provided to transport the boat between the entry load port and the assembler, where the boat is incorporated into an assembly between portions of the jig.

[0020] The assembler comprises a separator, which separates the received jig into a first jig portion and a second jig portion, and a loader that positions the boat between the first jig portion and the second jig portion to form the assembly. For example, the separator can comprise a hoist that is supported by a rail and travels along a length of the rail. The hoist cooperates with a top of the jig, for example, and lifts the top of the jig up from a base of the jig. The separator then travels along

the rail in a first direction to allow a loader to place the boat on top of the base of the jig, before the separator travels in an opposite direction to position the top of the jig above the boat resting on the base of the jig.

[0021] In some embodiments, the first robot receives the assembly prepared by the assembler. For example, the first robot can include an arm or other device that retrieves the assembly comprising the boat disposed between the top of the jig and the base of the jig from the assembler. According to some embodiments, the assembler delivers the assembly to the first robot, or a transport system is provided to convey the assembly between the assembler and the first robot.

[0022] Regardless of how the first robot receives or gains access to the assembly, the first robot comprises at least one mechanized tool that is operable under the control of electronic or computer-implemented control circuitry. In some embodiments, operation of the first robot manipulates a locking system of the assembly without manual user intervention to initiate or perform the manipulation of the locking system. For example, the manipulation described herein fixes an alignment of the boat relative to the first portion of the jig, the second portion of the jig, or both the first portion and the second portion of the jig.

[0023] To fix the relative arrangement of the portions of the jig and the boat, some embodiments of the assembly are transported by a conveyor belt or other transport system between the assembler and the first robot to bring the assembly within reach of the first robot. A mechanized tool is provided to the first robot to impart a compressive force on the boat between the top of the jig and base of the jig. For example, a torque wrench or other pivotal device exerts a rotational force at a defined torque on a threaded fastener forming part of a locking system provided to the assembly. In some embodiments, the torque wrench imparts a torque of at least one kilogram-foot (1 kg-ft) onto the threaded fastener or imparts a torque of between three kilogram-feet (3 kg-ft) and five kilogram-feet (5 kg-ft) onto the threaded fastener. The torque applied to the threaded fastener is chosen to be specific to the demands of the application.

[0024] According to some embodiments of the locking system, the threaded fastener is adjusted by the torque wrench to fix the alignment between the boat and at least one of the top of the jig or the base of the jig is described as one illustrative example. Other locking systems that are manipulated by the first robot to fix the relative arrangement of the boat and the at least one of the top of the jig or the base of the jig are also within the scope of the present disclosure. For example, any releasable fastener that can be manipulated by the first robot to fix or maintain the alignment between the boat and the first portion of the jig or the second portion of the jig during a processing operation and subsequently released can be utilized. Fixing the relative alignment between the boat and at least one portion of the jig with the locking system forms a locked assembly.

[0025] The locked assembly is then transported to a process chamber such as an oven, for example, where the locked chamber is subjected to an elevated temperature during a fabrication operation. In some embodiments, the second robot transfers the locked assembly to the process chamber. For example, the second robot places the locked assembly in the process chamber, deposits the locked chamber on a conveyor to be transported between the first robot and the process chamber, or otherwise sends the locked assembly to be placed within the process chamber.

[0026] In some embodiments, the locked assembly is subjected to the fabrication operation within the process chamber. An example of the process chamber is an oven, in which the locked assembly is subjected to an elevated temperature of at least one hundred Fahrenheit (100° F.). Another example of the process chamber is a deposition chamber, in which one or more layers are deposited on a substrate as part of a deposition process. The process chamber can be any chamber in which the locked assembly comprising the die(s) will be subjected to an operation as part of the process of manufacturing a semiconductor electronic device.

[0027] Once the operation performed in the process chamber is complete, some embodiments of the process involve returning the locked assembly to the first robot, which releases the threaded fastener or other locking system, without manual user intervention, to form an unlocked assembly.

The unlocked assembly is transported to the assembler. In some embodiments, the assembler reverses the operations performed to insert the boat between the first portion and the second portion of the jig to form the assembly, thereby separating the boat from the unlocked assembly.

[0028] For example, some embodiments of the assembler lift the top of the jig from the unlocked assembly, exposing the boat. In some embodiments, the loader removes the boat from the base of the jig and causes the removed boat to be transported to an exit port, from where the boat is removed from the system. For example, a second conveyor can be provided to transport the boat removed from the jig by the assembler to the exit port.

[0029] With reference to the drawings, FIG. 1 illustrates a system **100** for manufacturing a semiconductor electronic device in accordance with some embodiments. As shown, the system **100** comprises at least one jig carrier **102**, or a plurality of jig carriers **102**, comprising a plurality of storage cells **104**. The storage cells **104** are arranged in a configuration that allows the individual storage cells **104** to be assigned unique addresses. For example, as shown in FIG. 1, the storage cells **104** are arranged in rows and columns, with each storage cell **104** being identifiable by a respective intersecting row and column designator.

[0030] Some embodiments of the system **100** comprise an entry load port **106** through which a boat **300** (FIG. 3) supporting at least one die **302** is to be introduced to the system **100**. The entry load port **106** is configured to receive a boat magazine **108** manually transported by a technician, transported by an overhead transport system, or transported by any other mode of transportation. For example, the entry load port **106** can be a mobile launch platform (“MLP”) comprising a partition **110** defining an aperture **112** opening in an upward direction to allow the boat magazine **108** to be lowered by the overhead transport system and received by the entry load port **106**. A boat magazine **108** stores a plurality of boats **300**, each supporting one die **302** or a plurality of dies **302** that form portions of semiconductor electronic devices being fabricated during the fabrication process. The boat magazine **108** and boats **300** allow the die(s) **302** to be transported from one location to another as a batch.

[0031] For the embodiment illustrated in FIG. 1, the entry load port **106** is operatively coupled to an assembler **114**. Being operatively coupled to the assembler **114** allows a conveyor **116**, shuttle, or other transport system integrated as part of the entry load port **106** to deliver the boats **300** removed from the boat magazine **108** to the assembler **114** directly, without the involvement of an intervening or external transport device or human technicians, for example.

[0032] The jig carrier **102** and the entry load port **106** serve as sources of the components to be combined into an assembly **500** (FIG. 5) that is to be subjected to a fabrication operation by the system **100**. According to some embodiments, and as described in greater detail below, the assembler **114** comprises a separator **602** (FIG. 6) that separates a jig **200** (FIG. 2) into a first jig portion **202** (FIG. 2) and a second jig portion **204**. A loader **608** (FIG. 6) positions a boat **300** adjacent to the first jig portion **202** before the second jig portion **204** is placed above the boat **300** to form the assembly **500** comprising the boat **300** arranged between the first jig portions **202** and the second jig portion **204**.

[0033] The assembly **500** is to be subjected to a fabrication operation within a process chamber **122** defined by a fabrication device **124**. According to some embodiments, the fabrication device **124** is an oven, and the process chamber **122** (shown by dashed lines) is an oven cavity in which an elevated temperature of at least one hundred Fahrenheit (100° F.) is to be established. However, the fabrication device **124** is not limited to an oven. Some embodiments of the process chamber include, but are not limited to, a deposition chamber in which one or more layers are deposited on a substrate as part of a deposition process, an etching chamber in which a portion of a layer is etched away or otherwise removed, and the like. The process chamber **122** can be any chamber in which the assembly **500** is to be subjected to an operation as part of the process of manufacturing a semiconductor electronic device.

[0034] The system **100** comprises an automated material handling system to transport jigs **200** and

boats **300** as described herein to control or reduce the manual intervention and transport (e.g., by human hand) between the introduction of components to be assembled and a time when the boat **300** exits an exit port **126**. For example, a first robot **128** comprises a plurality of articulated arms **130**. The articulated arms **130** are electronically actuated, driven by electric stepper motors or other actuators. The articulated arms **130** can comprise a plurality of axes, or degrees of freedom, to position a tool **132** adjacent to a distal end of the articulated arms **130** in positions to manipulate a locking system **206** (FIG. 2) that fixes the relative positions of the boat **300** and the first jig portion **202** and the second jig portion **204** as described below.

[0035] According to some embodiments, a second robot **134** comprises an articulated arm **136** comprising six axes of rotation or six degrees of freedom. The articulated arm **136** of the second robot **134** comprises a clamp **138** or other receiver that is compatible with the jig **200** to retrieve the jig **200** from the jig carrier **102** storing the inventory of available jigs. The articulated arm **136** of the second robot **134** is electronically actuated, driven by an electric stepper motor or other actuators. The actuators of the first robot **128** and the second robot **134** are independently controllable by a computer-implemented control system configured as a computing device **1112** (FIG. 11), for example, to automatically transport the jig **200**, the boat **300**, or a combination thereof automatically, without manual user intervention, for example. The above descriptions of the first robot **128** and the second robot **134** are merely illustrative, and not exhaustive of the constructions within the scope of the present disclosure.

[0036] According to some embodiments, the exit port **126** comprises a conveyor **140** that is operable to transport a boat **300** from the system **100** following completion of the fabrication operation on the die **302**. Examples of the conveyor **140** include, but are not limited to, a conveyor belt, shuttle, roller table, or other device that receives a boat delivered by an adjacent conveyor and transports the boat **300** to an ambient environment of the system **100** through the exit port **126**. The boat **300** can be manually retrieved and transported away from the exit port **126**, retrieved by an overhead transport system, or otherwise removed from the exit port **126**. In some embodiments, the exit port **126** is an MLP and comprises a partition **142** defining an aperture **144** through which an overhead transport system retrieves the boat **300** or the boat magazine **108** from the exit port **126**.

[0037] A perspective view of a jig **200** in accordance with some embodiments is shown in FIG. 2. As shown, the jig **200** comprises a first jig portion **202** forming a base of the jig **200** and a second jig portion **204** that is supported on top of the first jig portion **202**. According to some embodiments, the second jig portion **204** is coupled to the first jig portion **202** by a locking system **206**, shown in FIG. 2 as comprising a plurality of threaded fasteners **208**, such as screws, bolts, etc., for example. A distance separating the second jig portion **204** from the first jig portion **202** is established by suitably manipulating the locking system **206** for the separation desired.

[0038] For example, the threaded fasteners **208** are pivotally adjustable in a first angular direction to insert the threaded fasteners **208** into compatibly threaded apertures to urge the second jig portion **204** toward the first jig portion **202**. The threaded fasteners **208** are pivotally adjustable in a second angular direction, which is the opposite of the first angular direction, for example, to remove the threaded fasteners **208** from the compatibly threaded apertures, thereby releasing the locking system **206**.

[0039] According to some embodiments, the tool **132** adjacent to the distal end of the articulated arms **130** is used to manipulate the locking system **206** to fix the alignment of the boat **300** relative to the first jig portion **202** and the second jig portion **204**, without manual intervention, for example, to form a locked assembly **500** as described herein. For example, the embodiment of the tool **132** shown in FIG. 1 comprises a motor **146** that drives a bit **148**. The bit **148** comprises a head that is compatible with a corresponding feature of the threaded fastener **208**, transmitting angular movement of the bit **148** to the threaded fastener **208**, thereby pivotally adjusting the threaded fasteners **208** of the jig **200**. According to some embodiments, the head comprises a socket, a flat blade, a cross shape formed by orthogonal blades, a star shape, a hexagonal post, etc. At least a

portion of the bit **148**, such as the head, is formed from a metallic material comprising a Rockwell hardness of 60 HRC or less. Hardness values of 60 HRC or less mitigate the likelihood of damaging the threaded fasteners **208** while manipulating the locking system **206** with the torque described herein.

[0040] According to some embodiments, the tool **132** comprises a torque controlling module. The torque controlling module can be implemented as part of the computing device **1112** to control electric energy supplied to the motor **146** and apply a defined torque to the threaded fastener **208**. The torque applied to the threaded fastener **208** can be correlated by the torque controlling module to apply a desired compressive force on the boat **300** positioned between the first jig portion **202** and the second jig portion **204**. In some embodiments, the torque applied can be at least one kilogram-foot (1 kg-ft), up to kilogram-feet five (5 kg-ft). In some embodiments, the applied force is between three kilogram-feet (3 kg-ft) and five kilogram-feet (5 kg-ft). The torque applied to the threaded fastener **208** controls the compressive force exerted between the first jig portion **202** and the second jig portion **204**. The compressive force urges an object, such as a heat sink, for example, separated from a surface of the die **302** by the thermal paste toward the die **302**. Depending on the type and quantity of thermal paste applied to the surface of the die **302**, the thermal paste is caused to spread over a substantial portion (e.g., at least 90%) of the surface of the die **302**, or the entirety of the surface of the die **302** as a result of the compressive force, without damaging the die **302** to an extent that renders the semiconductor device inoperable.

[0041] In some embodiments, the locking system **206** is manipulated to establish a desired height of the assembly **500** comprising the boat **300** disposed between the first jig portion **202** and the second jig portion **204**. The height of the assembly **500** is a dimension of the assembly **500** along an axis that is orthogonal to a major plane of the first jig portion **202**, second jig portion **204**, and boat **300**. According to some embodiments, the locking system **206** is adjustable to establish a height of the assembly to be thirty millimeters (30 mm) or less, or twenty nine and seven tenths millimeters (29.7 mm) or less, or twenty five millimeters (25 mm) or less, or the like. However, other values of the height can be established within the scope of the present disclosure to meet the specific demands of a given application. For example, the height of the assembly **500** can be established to position an object, such as a heat sink, for example, in close proximity to a surface of the die **302** supporting thermal paste. Compression of the thermal paste in the small space between the object and the surface of the die **302** causes the thermal paste to spread over a surface of the die **302** supported by the boat **300**, for example.

[0042] FIG. 3 is a top view of a boat **300** supporting a plurality of dies **302** in accordance with some embodiments. The boat **300** comprises a frame **304** defining a plurality of receptacles **306** that separately receive different dies **302**. A plurality of the boats **300** are received and transported collectively in a boat magazine **108**, such as that illustrated in FIG. 4.

[0043] According to some embodiments, lateral side walls **308** (FIG. 3) of the boats **300** are received in apertures **402** (FIG. 4) formed in lateral side walls **404** of the boat magazine **108**. According to some embodiments, the apertures **402** extend longitudinally, along a substantial extent of the lateral side walls **404**, and opposing pairs of the apertures **402** are formed at a plurality of different vertical elevations along the lateral side walls **404** to receive opposite lateral side walls **308** of the boat **300**. A plurality of boats **300** can be independently supported by a flange defining a bottom periphery of the apertures **402** formed in the lateral side walls **404**. Accordingly, the boats **300** in the boat magazine **108** are vertically arranged one above another and spaced apart from each other within the boat magazine **108**. In some embodiments, a head **406** extends upward from a ceiling **408** of the boat magazine **108**. The head **406** is compatible with a receiver provided to the overhead transport system, allowing the boat magazine **108** to be transported and introduced to the entry load port **106** and retrieved and transported from the exit port **126** by the overhead transport system, for example.

[0044] An illustrative example of the assembly **500** comprising the boat **300** disposed between the

first jig portion **202** and the second jig portion **204** is shown in FIG. 5, in accordance with some embodiments. The position of the boat **300** relative to the first jig portion **202** or the second jig portion **204**, or relative to both the first jig portion **202** and the second jig portion **204** is fixed, or otherwise maintained by cooperation between the threaded fasteners **208**, the boat **300**, and one or both of the first jig portion **202** and the second jig portion **204**.

[0045] An illustrative embodiment of an assembler **114** is shown in FIG. 6. As shown, the assembler **114** comprises a first platform **600** that supports the jig **200** during the formation and disassembly of the assembly **500**. A separator **602** which, according to some embodiments comprises a hoist **604** or other lifting device, separates the jig **200** into the first jig portion **202** and the second jig portion **204** by lifting the second jig portion **204** from the first jig portion **202**. A linear actuator **606** or other actuator such as a hydraulic or pneumatic cylinder, for example, is operable to elevate and lower the second jig portion **204** relative to the first jig portion **202**.

[0046] In some embodiments, the assembler **114** comprises a loader **608** that positions the boat **300** adjacent to the first jig portion **202** supported by the first platform **600**. For example, the loader **608** shown in FIG. 6 comprises a hoist **610** or other lifting device that can be elevated and lowered relative to the first jig portion **202** on the first platform **600** by a linear actuator **612** or other actuator such as a hydraulic or pneumatic cylinder, for example. When the boat **300** is suspended vertically above the first jig portion **202** supported by the first platform **600**, the linear actuator **612** is operable to lower the boat **300** onto the first jig portion **202**. The hoist **610** releases the boat **300**, thereby depositing the boat **300** onto the first jig portion **202**.

[0047] According to some embodiments, the assembler **118** comprises a second platform **614**. The second platform **614** is arranged adjacent to the conveyor **116** of the entry load port **106** to receive the boat **300** being introduced to the system **100**. At least one of the first platform **600** and second platform **614** or the separator **602** and loader **608** are movable in longitudinal directions of the assembler **118**, indicated generally by arrows **616**. For example, the position of the first platform **600** is adjustable along a rail **618**, or the separator **602** or the loader **608** are adjustable along a frame **620** to establish alignment between the first platform **600** or the second platform **614**, and the separator **602**, loader **608**, or any other portion of the assembler **114**. Alignment of the separator **602** with the first platform **600**, for example, involves positing the separator **602** so the hoist **604** is arranged vertically above the first platform **600**.

[0048] An illustrative example of a transport system **700** that transports at least one of the first jig portion **202**, the second jig portion **204**, the boat **300**, or a combination thereof within the system **100** is shown in FIG. 7. In some embodiments, the transport system **700** forms a portion of the conveyor **116**, the first platform **600**, the second platform **614**, or any other conveyor or device described herein as transporting at least one of the first jig portion **202**, the second jig portion **204**, the boat **300**, or a combination thereof within the system **100**. According to some embodiments, the transport system **700** comprises a track **702**. The track **702** comprises a continuous belt **704** that extends around a plurality of pulleys **706**. At least one of the pulleys **706** is operatively connected to a drive motor **708**. Activation of the drive motor **708** in a first direction causes the continuous belt **704** to travel in a first direction about the pulleys **706**, and activation of the drive motor **708** in a second direction causes the continuous belt **704** to travel in a second direction about the pulleys **706**. The second direction the continuous belt **704** travels is opposite to the first direction in which the continuous belt **704** travels.

[0049] The continuous belt **704** traveling in the first direction and the second direction transports at least one of the first jig portion **202**, the second jig portion **204**, the boat **300**, or an assembly comprising a combination of the boat **300**, the first jig portion **202**, and the second jig portions **204** in different directions across the track **702**. In some embodiments, the transport system **700** comprises a plurality of tracks **702**, laterally spaced apart from each other as shown in FIG. 7, supporting the object being transported using the plurality of tracks **702**. According to such embodiments, the drive motor **708** is operatively connected to one or more pulleys **706** provided to

each of the tracks **702** and is operable to cause the continuous belt **704** of each track **702** to travel in one or more directions about the pulleys **706**. According to other embodiments, a separate drive motor **708** is provided to independently drive the continuous belt **704** of the tracks **702** about the pulleys **706**, or a drive motor **708** is provided to a first one of the plurality of tracks **702**, and not to a second one of the plurality of tracks **702**, to drive the continuous belt **704** of one of the tracks **702** but not the other.

[0050] According to some embodiments, the system **100** comprises an arrangement of transport systems **700** relative to the first robot **128** and the second robot **134**, as illustrated in FIG. **8** that allows the first robot **128** and the second robot **134** to be utilized at a plurality of different times during a computer-implemented method of manufacturing a semiconductor electronic device. For example, a pre-press conveyor **800** is disposed between the first robot **128** and the second robot **134**. In such a position, the pre-press conveyor **800** is accessible by both the first robot **128** and the second robot **134**, allowing the first robot **128** and the second robot **134** to interact with objects on the pre-press conveyor **800**, for example.

[0051] The second robot **134** is operable to transfer the jig **200** between the jig carrier **102** and the pre-press conveyor **800** and to return the jig **200** from the pre-press conveyor **800** to the jig carrier **102**. In some embodiments, the second robot **134** is located adjacent to, or between, the jig carrier **102** and the fabrication device **124**. From such a location, the second robot **134** transfers the jig **200** between the jig carrier **102** and the pre-press conveyor **800** and between the pre-press conveyor **800** and the fabrication device **124**.

[0052] In some embodiments, a buffer station **802** is arranged adjacent to a first end of the pre-press conveyor **800** to receive the assembly **500** comprising the boat **300** arranged between the first jig portion **202** and the second jig portion **204**. An illustrative embodiment of the buffer station **802** comprises the transport system **700** to transport the assembly **500**. The buffer station **802** supports the assembly **500** at least one of before or after the assembly **500** is subjected to the fabrication operation in the process chamber **122**, as needed, to control or reduce interruptions to the fabrication process. For example, an assembly **500** to be subjected to the fabrication operation is to be received by the buffer station **802**. The pre-press conveyor **800** is operated to convey the assembly **500** to the buffer station **802**, which is also operated, causing the assembly **500** to roll off of the tracks **702** of the pre-press conveyor **800** and onto the tracks **702** of the buffer station **802**.

[0053] In some embodiments, the buffer station **802** is not accessible by the second robot **134**. The buffer station **802** is pivotally adjustable about a central axis **804** in directions indicated by arrows **806**. A portion of the assembly **500** on the buffer station **802** that is not accessible by the second robot **134** while the buffer station **802** is in a first angular orientation about the central axis **804**, is accessible by the second robot **134** while the buffer station **802** is in a second angular orientation, different than the first angular orientation, about the central axis **804**. For example, the buffer station **802** is oriented with a longitudinal axis parallel with a longitudinal axis of the pre-press conveyor **800** in a first orientation. The buffer station **802** is pivoted in a counter-clockwise direction a defined angle about the central axis **804** to the second orientation. The defined angle is any angle between fifteen degrees ( $15^\circ$  and one hundred sixty five degrees ( $165^\circ$ , however, some embodiments of the defined angle that the buffer station **802** is pivotally adjustable from zero degrees ( $0^\circ$  to three hundred sixty degrees ( $360^\circ$ . The angle that the buffer station **802** can be pivotally adjusted to the second angular orientation about the central axis **804** of the buffer station **802** is suitable to place the assembly **500** on the buffer station **802** within reach of the second robot **134**. With the buffer station **802** adjusted to such a position, the second robot **134** is able to retrieve the assembly **500** from the buffer station **802**, and introduce the assembly **500** to the process chamber **122**.

[0054] In some embodiments, the second robot **134** transfers the assembly **500** from the process chamber **122** to the buffer station **802** after completion of the fabrication operation. For example, if the pre-press conveyor **800** is in use, the buffer station **802** is pivotally adjusted to the second

angular orientation about the central axis **804**. The second robot **134** retrieves the assembly **500** from the process chamber **122** and introduces the assembly **500** to the buffer station **802** in the second angular orientation. The buffer station **802** is pivotally adjusted to the first angular orientation about the central axis **804**. When the pre-press conveyor **800** is no longer in use, or is available to receive the assembly **500** from the buffer station **802**, the drive motor **708** of the buffer station **802** and the drive motor **708** pre-press conveyor **800** are activated, conveying the assembly **500** from the buffer station **802** to the pre-press conveyor **800**.

[0055] Some embodiments of the system **100** utilize the jig carrier **102** to provide added buffer capacity or to provide storage capacity for the assembly **500** to cool after being exposed to an elevated temperature within the process chamber **122**, in addition to storing the inventory of jigs **200**. For example, the second robot **134** places the assembly **500**, locked or unlocked, in a storage cell **104** of the jig carrier **102** to control or reduce the number of interruptions in the process caused by two or more objects being present at the same location within the system **100** at the same time. In some embodiments, the second robot **134** introduces the assembly **500** to a storage cell **104** to cool after being exposed to an elevated temperature in the process chamber **122**, or otherwise as needed.

[0056] In some embodiments, the pre-press conveyor **800** is also accessible by the first robot **128**. For such embodiments, the tool **132** provided adjacent to the distal end of each articulated arm **130** may manipulate the locking system **206** to maintain a relationship between the first jig portion **202** relative to the second jig portion **204** delivered to the first robot **128** by the pre-press conveyor **800** before performance of the fabrication operation. After the fabrication operation, the tool **132** provided adjacent to the distal end of each articulated arm **130** releases the locking system **206**, for example.

[0057] According to some embodiments, a port conveyor **808** transports at least the boat **300** between the assembler **114** and the exit port **126**. For example, the port conveyor **808** comprises the transport system **700** comprising one or a plurality of tracks **702** extending continuously between the assembler **114** and the exit port **126**. The drive motor **708** of the port conveyor **808** is operable to cause the continuous belt **704** of the port conveyor **808** to travel about pulleys **706**, transferring the boat **300** being conveyed from the second platform **614** of the assembler **114**. In some embodiments, the port conveyor **808** extending between the assembler **114** and the exit port **126** is separated from a portion of the pre-press conveyor **800** by the first robot **128**. For such embodiments, the entry load port **106**, the port conveyor **808**, and the exit port **126** form a boat side of the system **100**, which does not receive the jig **200** as part of the manufacturing process.

[0058] A flow chart illustrating a method of manufacturing a semiconductor electronic device in accordance with some embodiments is shown in FIG. **9**. In some embodiments, the actions illustrated in FIG. **9** are performed under the control of a computer-implemented control system, such as computing device **1112** in FIG. **11** for example, executing computer-executable instructions stored by a non-transitory, computer-readable medium **1008**, shown in FIG. **10**. According to some embodiments, the method comprises forming the assembly **500** comprising the first jig portion **202**, the second jig portion **204**, and the boat **300** supporting a die, at block **900**. The boat **300** is arranged between the first jig portion **202** and the second jig portion **204**. More specifically, in some embodiments the second robot **134** is controlled to retrieve the jig **200** from the jig carrier **102** and deliver the jig **200** to the pre-press conveyor **800**. The drive motor **708** of the pre-press conveyor **800** and the drive motor **708** of the first platform **600** are activated to transfer the jig **200** from the pre-press conveyor **800** to the assembler **114**.

[0059] In some embodiments, the assembler **114** separates the second jig portion **204** from the first jig portion **202**, places the boat **300** in a stack comprising the first jig portion **202**, and places the second jig portion **204** on the assembly **500**. For example, the relative position of the separator **602** and the first platform **600** is adjusted to position the hoist **604** vertically above the second jig portion **204**. The first platform **600**, the separator **602**, or both the first platform **600** and the

separator **602** are caused to travel parallel to the arrows **616** to establish vertical alignment between the hoist **604** and the second jig portion **204**. When the separator **602** is arranged vertically above the first platform **600**, the hoist **604** is lowered by the linear actuator **606**, causing the hoist **604** to cooperate with the second jig portion **204**, and then raised by the linear actuator **606** to lift the second jig portion **204** from the first jig portion **202**.

[0060] The boat **300** retrieved from the boat magazine **108** that has been delivered to the entry load port **106** by an overhead transport system is transported by the conveyor **116** of the entry load port **106** to the second platform **614** of the assembler **114**. The second platform **614**, the loader **608**, or both the second platform **614** and the loader **608** are caused to travel parallel to the arrows **616** to establish vertical alignment between the hoist **610** of the loader **608** and the boat **300** on the second platform **614**. When the loader **608** is arranged vertically above the boat **300**, the hoist **610** is lowered by the linear actuator **612**, causing the hoist **610** to cooperate with the boat **300**, and then raised by the linear actuator **612** to lift the boat **300** from the second platform **614**. The loader **608** and the first jig portion **202** supported by the first platform **600** are brought into vertical alignment, and the hoist **610** coupled to the boat **300** is lowered by the linear actuator **612** to place the boat **300** on top of the first jig portion **202**. Relative alignment of the separator **602** vertically above the second jig portion **204** is established, and the linear actuator **606** lowers the hoist **604** to deposit the second jig portion **204** onto the boat **300** supported by the first jig portion **202** on the first platform **600** to form the assembly **500**.

[0061] In some embodiments, a locked assembly **500** is formed by the first robot **128** at block **902**. For example, the assembly **500** is delivered to the first robot **128** as a result of activating the drive motor **708** of the first platform **600** and the drive motor **708** of the pre-press conveyor **800**. The drive motor **708** of the pre-press conveyor **800** remains operational to position the assembly **500** adjacent to the first robot **128**, within reach of the articulated arms **130**, allowing the first robot **128** to manipulate the locking system **206** using the tool **132** provided to each of the articulated arms **130** without manual user intervention, for example. In some embodiments, the tools **132** are governed by the torque controlling module implemented as computer-executable instructions, to cause the tools to pivotally adjust threaded fasteners **208** with a torque of at least one kilogram-foot (1 kg-ft) onto the threaded fasteners **208**. In some embodiments the applied torque is between three kilogram-feet (3 kg-ft) and five kilogram-feet (5 kg-ft).

[0062] In some embodiments, the second robot **134** is operable to transport the locked assembly **500** to the process chamber **122** of the fabrication device **124**, where the locked assembly **500** is to be subjected to a processing operation. The locked assembly **500** is retrieved by the second robot **134** from the pre-press conveyor **800** and introduced to the process chamber **122**. If there is an interruption in the process, the locked assembly **500** is transported to the buffer station **802** before being transported to the process chamber **122**. The buffer station **802** is pivotally adjustable to position the locked assembly **500** within reach of the second robot **134**, allowing the second robot **134** to retrieve the locked assembly **500** from the buffer station **802** and transport the locked assembly **500** to the process chamber **122**.

[0063] Once in the process chamber **122**, the locked assembly **500** is subjected to a fabrication operation at block **904**. An example of the fabrication device **124** includes an oven chamber as the process chamber **122**, in which the locked assembly **500** is exposed to an elevated temperature of at least one hundred Fahrenheit (100° F.). Other examples of the process chamber **122** include a deposition chamber in which a material is deposited on the die **302** supported by the boat **300**, a test chamber in which a test is performed on the die **302**, an etching chamber in which a portion of the die **302** is removed, etc.

[0064] Following completion of the fabrication operation, the second robot **134** retrieves the locked assembly **500** and transports the locked assembly **500** between the process chamber **122** and the first robot **128** at block **906**. The second robot **134** transports the locked assembly **500** to the pre-press conveyor **800** or, if there is an interruption in the process, to the buffer station **802** until a time

when the interruption has been resolved. For example, the interruption includes the pre-press conveyor **800** being occupied by another assembly **500** to be transported to the fabrication device **124** or another locked assembly **500** being unlocked before being transported back to the assembler **114**.

[0065] In some embodiments, the first robot **128** again manipulates the locking system **206** at block **908** to release the locking system **206** and form an unlocked assembly **500** that is transported to the assembler **114**. For example, the first robot **128** forms the unlocked assembly **500** by pivotally adjusting the threaded fasteners **208** using the tools **132**, without manual intervention, for example. The drive motor **708** of the pre-press conveyor **800** and the drive motor **708** of the first platform **600** are operated to return the unlocked assembly **500** to the assembler **114**.

[0066] The unlocked assembly **500** is returned to the first platform **600** of the assembler **114**, where the unlocked assembly **500** is disassembled by separating the boat **300** from the first jig portion **202** and the second jig portion **204** at block **910**. For example, relative alignment between the separator **602** and the unlocked assembly **500** is established by adjusting the position of at least one of the separator **602** or the first platform **600** in directions parallel with arrows **616**. The linear actuator **606** is activated to lower the hoist **604** and engage the second jig portion **204**. The linear actuator **606** of the separator **602** is controlled to elevate the hoist **604** and lift the first jig portion **202** from the boat **300**. The loader **608** is brought into alignment with the first platform **600**, and the linear actuator **606** controls the elevation of the hoist **610** to lift the boat **300** from the first jig portion **202**.

[0067] In some embodiments, the loader **608** is brought into alignment with the second platform **614**, and the linear actuator **612** is controlled to deposit the boat **300** onto the second platform **614**. The drive motor **708** of the second platform **614** and the drive motor **708** of the port conveyor **808** are activated, transferring the boat **300** to the port conveyor **808**. The port conveyor **808** is operated to transport the boat **300** to the exit port **126**, where the boat **300** is introduced to a boat magazine **108** to be retrieved and transported away from the system **100** by an overhead transport system.

[0068] Still another embodiment involves a computer-readable medium comprising processor-executable instructions configured to implement one or more of the techniques presented herein. An exemplary computer-readable medium **1008** is illustrated in FIG. **10**, and is embodied as a CD-R, DVD-R, flash drive, a platter of a hard disk drive, etc., on which is encoded computer-readable data **1006**. This computer-readable data **1006** in turn comprises a set of processor-executable computer instructions **1004** configured to operate according to one or more of the principles set forth herein. In some embodiments, the processor-executable computer instructions **1004** are configured to perform a method **1002**, such as at least some of the aforementioned described methods. In some embodiments, the processor-executable computer instructions **1004** are configured to implement a system, such as at least some of the aforementioned systems. Many such computer-readable media may be devised by those of ordinary skill in the art that are configured to operate in accordance with the techniques presented herein.

[0069] FIG. **11** and the following discussion provide a brief, general description of a suitable computing environment to implement embodiments of one or more of the provisions set forth herein. The operating environment of FIG. **11** is only one example of a suitable operating environment and is not intended to suggest any limitation as to the scope of use or functionality of the operating environment. Example computing devices include, but are not limited to, personal computers, server computers, hand-held or laptop devices, mobile devices (such as mobile phones, Personal Digital Assistants (PDAs), media players, and the like), multiprocessor systems, consumer electronics, mini computers, mainframe computers, distributed computing environments that include any of the above systems or devices, and the like.

[0070] Although not required, embodiments are described in the general context of “computer readable instructions” being executed by one or more computing devices. Computer readable instructions may be distributed via computer readable media (discussed below). Computer readable

instructions may be implemented as program modules, such as functions, objects, Application Programming Interfaces (APIs), data structures, and the like, that perform particular tasks or implement particular abstract data types. Typically, the functionality of the computer readable instructions may be combined or distributed as desired in various environments.

[0071] FIG. **11** depicts an example of a system **1100** comprising a computing device **1112** configured as the controller system to implement some embodiments provided herein. In some configurations, computing device **1112** includes at least one processing unit **1116** and memory **1118**. Depending on the exact configuration and type of computing device, memory **1118** may be volatile (such as RAM, for example), non-volatile (such as ROM, flash memory, etc., for example) or some combination of the two. This configuration is illustrated in FIG. **11** by dashed line **1114**.

[0072] In some embodiments, computing device **1112** may include additional features and/or functionality. For example, computing device **1112** may also include additional storage (e.g., removable and/or non-removable) including, but not limited to, magnetic storage, optical storage, and the like. Such additional storage is illustrated in FIG. **11** by storage **1120**. In some embodiments, computer readable instructions to implement one or more embodiments provided herein may be in storage **1120**. Storage **1120** may also store other computer readable instructions to implement an operating system, an application program, and the like. Computer readable instructions may be loaded in memory **1118** for execution by processing unit **1116**, for example.

[0073] The term “computer readable media” as used herein includes computer storage media. Computer storage media includes volatile and nonvolatile, removable and non-removable media implemented in any method or technology for storage of information such as computer readable instructions or other data. Memory **1118** and storage **1120** are examples of computer storage media. Computer storage media includes, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, Digital Versatile Disks (DVDs) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by computing device **1112**. Any such computer storage media may be part of computing device **1112**.

[0074] Computing device **1112** may also include communication connection(s) **1126** that allows computing device **1112** to communicate with other devices. Communication connection(s) **1126** may include, but is not limited to, a modem, a Network Interface Card (NIC), an integrated network interface, a radio frequency transmitter/receiver, an infrared port, a USB connection, or other interfaces for connecting computing device **1112** to other computing devices. Communication connection(s) **1126** may include a wired connection or a wireless connection. Communication connection(s) **1126** may transmit and/or receive communication media.

[0075] The term “computer readable media” may include communication media. Communication media typically embodies computer readable instructions or other data in a “modulated data signal” such as a carrier wave or other transport mechanism and includes any information delivery media. The term “modulated data signal” may include a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal.

[0076] Computing device **1112** may include input device(s) **1124** such as keyboard, mouse, pen, voice input device, touch input device, infrared cameras, video input devices, and/or any other input device. Output device(s) **1122** such as one or more displays, speakers, printers, and/or any other output device may also be included in computing device **1112**. Input device(s) **1124** and output device(s) **1122** may be connected to computing device **1112** via a wired connection, wireless connection, or any combination thereof. In some embodiments, an input device or an output device from another computing device may be used as input device(s) **1124** or output device(s) **1122** for computing device **1112**.

[0077] Components of computing device **1112** may be connected by various interconnects, such as a bus. Such interconnects may include a Peripheral Component Interconnect (PCI), such as PCI Express, a Universal Serial Bus (USB), firewire (IEEE 1394), an optical bus structure, and the like.

In some embodiments, components of computing device **1112** may be interconnected by a network. For example, memory **1118** may be comprised of multiple physical memory units located in different physical locations interconnected by a network.

[0078] Those skilled in the art will realize that storage devices utilized to store computer readable instructions may be distributed across a network. For example, a computing device **1130** accessible via a network **1128** may store computer readable instructions to implement one or more embodiments provided herein. Computing device **1112** may access computing device **1130** and download a part or all of the computer readable instructions for execution. Alternatively, computing device **1112** may download pieces of the computer readable instructions, as needed, or some instructions may be executed at computing device **1112** and some at computing device **1130**.

[0079] In some embodiments, configuring the system **100** to position the first robot **128** and the second robot **134** adjacent to the jig carrier **102**, the assembler **114**, the fabrication device **124**, and the transport systems **700** controls or reduces manual intervention required to fabricate a semiconductor electronic device. Additionally, efficient use of the first robot **128** and the second robot **134** is made at various points during the manufacturing process to control or reduce interruptions to the process and control or reduce the need for separate material handling systems, for example.

[0080] According to some embodiments, a system for manufacturing a semiconductor electronic device is provided herein. The system includes an assembler configured to receive a jig and a boat supporting a die. The assembler includes a separator configured to separate the jig into a first jig portion and a second jig portion and a loader configured to position the boat adjacent to the first jig portion. The separator is configured to position the second jig portion adjacent to the boat to form an assembly including the boat arranged between the first jig portion and the second jig portion. A robot is configured to receive the assembly prepared by the assembler and manipulates a locking system that fixes an alignment of the boat relative to the first jig portion and the second jig portion to form a locked assembly. A process chamber is configured to receive the locked assembly and subjects the locked assembly to a fabrication operation.

[0081] According to some embodiments, a method of manufacturing a semiconductor electronic device is provided herein. The method includes forming an assembly including a first jig portion, a second jig portion, and a boat supporting a die arranged between the first jig portion and the second jig portion. With a first robot, a locked assembly is formed by manipulating a locking system to urge the first jig portion toward the second jig portion and fix an alignment of the boat relative to the first jig portion and the second jig portion. The locked assembly is subjected to a fabrication operation within a process chamber, before the first robot forms an unlocked assembly by releasing the locking system. The boat is separated from the unlocked assembly.

[0082] According to some embodiments, a non-transitory computer-readable medium storing instructions that, when executed by a processor of a computer-implemented control system, cause the computer-implemented control system to perform a process of manufacturing a semiconductor electronic device is provided herein. The manufacturing process includes retrieving, with a first robot, a jig from a jig carrier storing an inventory of available jigs. The jig includes a first jig portion and a second jig portion. The jig is delivered to an assembler, and a boat supporting a die is introduced to the assembler by an entry load port. The assembler arranges the boat between the first jig portion and the second jig portion to form an assembly, which is transported by a first conveyor to a second robot. The second robot manipulates a locking system of the assembly to form a locked assembly, thereby urging the first jig portion toward the second jig portion and exerting a compressive force on the boat. The first robot delivers the locked assembly to a processing chamber in which the locked assembly is to be subjected to a fabrication operation. The first robot returns the locked assembly to the second robot after completion of the fabrication operation, and the second robot releases the locking system to form an unlocked assembly. The unlocked assembly is transported to the assembler, which separates the boat from the unlocked assembly, and the boat

is delivered by a second conveyor to an exit port.

[0083] It will be appreciated that the features, elements, etc. are not limited to the specific configurations, compositions, arrangements, embodiments, etc. mentioned herein. In some embodiments, one or more of the configurations, compositions, arrangements, etc. mentioned herein comprise one or more structures, functionalities, etc. different than what is mentioned herein.

[0084] The foregoing outlines features of several embodiments so that those of ordinary skill in the art may better understand various aspects of the present disclosure. Those of ordinary skill in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes or achieving the same advantages of various embodiments introduced herein. Those of ordinary skill in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

[0085] Although the subject matter has been described in language specific to structural features or methodological acts, it is to be understood that the subject matter of the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed as example forms of implementing at least some of the claims.

[0086] Various operations of embodiments are provided herein. The order in which some or all of the operations are described, or some or all of the structures are arranged should not be construed to imply that these operations are necessarily order dependent. Alternative ordering will be appreciated having the benefit of this description. Further, it will be understood that not all operations are necessarily present in each embodiment provided herein. Also, it will be understood that not all operations are necessary in some embodiments.

[0087] It will be appreciated that the configurations, compositions, arrangements, features, arrangements, elements, etc. depicted herein are illustrated with particular dimensions relative to one another, such as structural dimensions or orientations, for example, for purposes of simplicity and ease of understanding and that actual dimensions of the same differ substantially from that illustrated herein, in some embodiments.

[0088] Moreover, “exemplary” or “illustrative” is used herein to mean serving as an example, instance, illustration, etc., and not necessarily as advantageous. As used in this application, “or” is intended to mean an inclusive “or” rather than an exclusive “or”. In addition, “a” and “an” as used in this application and the appended claims are generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form. Also, at least one of A and B and/or the like generally means A or B or both A and B. Furthermore, to the extent that “includes”, “having”, “has”, “with”, or variants thereof are used, such terms are intended to be inclusive in a manner similar to the term “comprising”. Also, unless specified otherwise, “first,” “second,” or the like are not intended to imply a temporal aspect, a spatial aspect, an ordering, etc. Rather, such terms are merely used as identifiers, names, etc. for features, elements, items, etc. For example, a first element and a second element generally correspond to element A and element B or two different or two identical elements or the same element.

[0089] Also, although the disclosure has been shown and described with respect to one or more implementations, equivalent alterations and modifications will occur to others of ordinary skill in the art based upon a reading and understanding of this specification and the annexed drawings. The disclosure includes all such modifications and alterations and is limited only by the scope of the following claims. In particular regard to the various functions performed by the above described components the terms used to describe such components are intended to correspond, unless otherwise indicated, to any component which performs the specified function of the described component (for example, a term that is functionally equivalent), even though not structurally equivalent to the disclosed structure. In addition, while a particular feature of the disclosure may

have been disclosed with respect to only one of several implementations, such feature may be combined with one or more other features of the other implementations as may be desired and advantageous for any given or particular application.

## Claims

1. A method of manufacturing a semiconductor electronic device, comprising: arranging a boat between a first jig portion and a second jig portion to form an assembly; performing a fabrication process on one or more die disposed within the boat while the one or more die are disposed between the first jig portion and the second jig portion; and separating the boat from the first jig portion and the second jig portion after performing the fabrication process.
2. The method of claim 1, comprising: introducing the boat through a load entry port prior to arranging the boat between the first jig portion and the second jig portion.
3. The method of claim 2, comprising: delivering the boat to an exit port after separating the boat from the first jig portion and the second jig portion.
4. The method of claim 1, comprising: locking the boat to at least one of the first jig portion or the second jig portion after arranging the boat between the first jig portion and the second jig portion.
5. The method of claim 4, comprising: unlocking the boat from the at least one of the first jig portion or the second jig portion after performing the fabrication process and before separating the boat from the first jig portion and the second jig portion.
6. The method of claim 4, wherein locking the boat to the at least one of the first jig portion or the second jig portion comprises: pivotally adjusting a threaded fastener to urge the second jig portion toward the first jig portion and exert a compressive force on the boat between the first jig portion and the second jig portion.
7. The method of claim 6, wherein pivotally adjusting the threaded fastener comprises pivotally adjusting the threaded fastener such that a defined torque is exerted by on the threaded fastener.
8. The method of claim 7, wherein the defined torque is at least one kilogram-foot.
9. The method of claim 7, wherein the defined torque is no greater than five kilogram-feet.
10. The method of claim 1, comprising: separating a jig into the first jig portion and the second jig portion prior to arranging the boat between the first jig portion and the second jig portion.
11. The method of claim 1, wherein the fabrication process is a heating process.
12. The method of claim 1, wherein the one or more die are present in the boat when arranging the boat between a first jig portion and a second jig portion.
13. A system for manufacturing a semiconductor electronic device, comprising: a lifting device configured to lift a first jig portion to separate the first jig portion from a second jig portion; and a loader configured to place a boat on top of the second jig portion while the first jig portion is separated from the second jig portion, wherein the lifting device is configured to position the first jig portion above the boat after the loader has placed the boat on top of the first jig portion to form an assembly of the first jig portion, the boat, and the second jig portion.
14. The system of claim 13, comprising: a rail on which the second jig portion moves between the lifting device and the loader.
15. The system of claim 13, comprising: a robot configured to impart a compressive force on the assembly.
16. The system of claim 15, wherein the robot is configured to exert a rotational force on a threaded fastener to couple the boat to at least one of the first jig portion or the second jig portion.
17. The system of claim 13, comprising: a robot configured to deliver the assembly to a processing chamber.
18. The system of claim 13, comprising: a locking system configured to compress the assembly such that a total height of the assembly is equal to a specified height.
19. An assembly, comprising: a first jig portion; a second jig portion; and a boat disposed between

the first jig portion and the second jig portion, wherein columns extend between the first jig portion and the second jig portion.

**20.** The assembly of claim 19, comprising: threaded fasteners to adjust a relative position between the first jig portion and the second jig portion.

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